SYSTEM ANALYSIS

Why Local AI Coding Agents Get Stuck in Apology Loops

How to fix JSON parse crashes and context eviction.

SLIDE 2

Separating the Brain from the Execution Layer

An agentic developer environment relies on two separate layers:

- **1. The Model Layer (Brain):** The LLM that processes prompts and decides **what** to do.
- **2. The Agent Runtime (Body):** The host framework (Cline, Continue) that manages filesystem tools and executes commands.

Failure occurs when output formats between these layers do not align.

SLIDE 3

General chat capability is different from tool execution.

Running standard models (like **Gemma 4 12B**) works for conversational coding queries, but fails when executing file edits. **The Reason:** If a model is not explicitly pre-trained or fine-tuned for native **function calling (tool-use)**, it will output text summaries instead of calling filesystem tools.

SLIDE 4

JSON formatting is fragile under open weights.

```
[Agent Runtime Parser Error]: Unexpected token 'S' in JSON

  [LLM Output]:
  Sure, let me help you edit that file. Here is the JSON:
  {
    "tool": "write_file",
    "arguments": { ... }
  }
```

Surrounding conversational padding breaks standard JSON parsers, causing infinite retries.

SLIDE 5

XML is more resilient than JSON.

The JSON Trap JSON is like a strict form: missing a single comma rejects the entire payload. Smaller local models struggle with strict syntax limits.

The XML envelope

```
<write_file path="main.ts">
  code
</write_file>
```

XML tags act like a bright red envelope. The parser spots the tags and extracts the code inside, ignoring all conversational padding around it.

SLIDE 6

Context eviction makes agents lose track.

Why do agents start outputting generic summaries mid-task?

Think of your context window as a **whiteboard**. As the conversation proceeds, the board fills up. Once full, you must erase the top lines to keep writing.

If you erase the original system instructions at the very top, the agent forgets its schemas and loses its tool behavior.

SLIDE 7

How to stabilize local agent loops:

- **Lower Temperature:** Set `temperature: 0.0 - 0.2` to enforce strict formatting.
    - **System Prompts:** Instruct the model to suppress conversational headers.
    - **Context Eviction:** Implement sliding window summarization to prevent tool execution history from overflowing the prompt window.

Read the full analysis on Medium.

Detailed metrics, XML parsing code blocks, and context management strategies are in the full article.

Link in Bio